FLAT PULLBACK OF THE CYCLE CLASS OF A CLOSED SUBSCHEME AND OF A COHERENT SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xypic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Dimension of the inverse image of a cycle under a flat morphism. Same chapter chow.tex, label lemma-flat-inverse-image-dimension, invoked at proof line 2353. Gives the two dimension statements immediately, since the support of the pulled back sheaf is the inverse image of the support.
- Cycle class of a closed subscheme versus that of its structure sheaf, and the structure sheaf of a preimage. Same chapter chow.tex, label lemma-cycle-closed-coherent, invoked at proof line 2354. Reduces part (1) to part (2) using that the pullback of the structure sheaf of Z is the structure sheaf of the preimage of Z.
- Coherence of a sheaf on a locally Noetherian scheme and of flat pullback, and the length comparison for a flat local extension. Cohomology of Schemes, label coherent-lemma-coherent-Noetherian (proof lines 2360 and 2364); Modules, label modules-lemma-pullback-finite-type (proof line 2362); Same chapter chow.tex, label lemma-length-finite (proof line 2376); Algebra, label algebra-lemma-pullback-module (proof line 2387). Makes the statement meaningful by showing a finite type sheaf on a locally Noetherian scheme is coherent and that flat pullback preserves this, and supplies the final equality relating length_B(B tensor_A M) to length_A(M) times length_B(B/m_A B).

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Source reference. Original lemma 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Navigation pointer for \ref{algebra-lemma-pullback-module} used inside the counted span (line 2387); the argument that consumes it is retained in the span.

Source reference. Original lemma 01XZ: sources/stacks-chow/tags.txt, source lines 2538–2538. Navigation pointer for \ref{coherent-lemma-coherent-Noetherian} used inside the counted span (line 2360); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QY: sources/stacks-chow/pinned-original-source.tex, source lines 1776–1776. Navigation pointer for \ref{lemma-cycle-closed-coherent} used inside the counted span (line 2354); the argument that consumes it is retained in the span.

Source reference. Original lemma 02R8: sources/stacks-chow/pinned-original-source.tex, source lines 2127–2127. Navigation pointer for \ref{lemma-flat-inverse-image-dimension} used inside the counted span (line 2353); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QW: sources/stacks-chow/pinned-original-source.tex, source lines 1702–1702. Navigation pointer for \ref{lemma-length-finite} used inside the counted span (line 2376); the argument that consumes it is retained in the span.

Source reference. Original lemma 01B6: sources/stacks-chow/tags.txt, source lines 1715–1715. Navigation pointer for \ref{modules-lemma-pullback-finite-type} used inside the counted span (line 2362); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 2332); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be locally of finite type over S. Let $f : X \to Y$ be a flat morphism of relative dimension r.*

1. *Let $Z \subset Y$ be a closed subscheme with $\dim_\delta(Z) \leq k$. Then we have $\dim_\delta(f^{-1}(Z)) \leq k + r$ and $[f^{-1}(Z)]_{k+r} = f^*[Z]_k$ in $Z_{k+r}(X)$.*
2. *Let $\mathcal{F}$ be a coherent sheaf on Y with $\dim_\delta(Supp(\mathcal{F})) \leq k$. Then we have $\dim_\delta(Supp(f^*\mathcal{F})) \leq k + r$ and*
$$f^*[\mathcal{F}]_k = [f^*\mathcal{F}]_{k+r}$$
in $Z_{k+r}(X)$.

Proof. The statements on dimensions follow immediately from Lemma 02R8. Part (1) follows from part (2) by Lemma 02QY and the fact that $f^*\mathcal{O}_Z = \mathcal{O}_{f^{-1}(Z)}$.

Proof of (2). As X, Y are locally Noetherian we may apply Cohomology of Schemes, Lemma 01XZ to see that $\mathcal{F}$ is of finite type, hence $f^*\mathcal{F}$ is of finite type (Modules, Lemma 01B6), hence $f^*\mathcal{F}$ is coherent (Cohomology of Schemes, Lemma 01XZ again). Thus the lemma makes sense. Let $W \subset Y$ be an integral closed subscheme of δ-dimension k, and let $W' \subset X$ be an integral closed subscheme of dimension $k + r$ mapping into W under f. We have to show that the coefficient n of $[W']$ in $f^*[\mathcal{F}]_k$ agrees with the coefficient m of $[W']$ in $[f^*\mathcal{F}]_{k+r}$. Let $\xi \in W$ and $\xi' \in W'$ be the generic points. Let $A = \mathcal{O}_{Y,\xi}$, $B = \mathcal{O}_{X,\xi'}$ and set $M = \mathcal{F}_\xi$ as an A-module. (Note that M has finite length by our dimension assumptions, but we actually do

not need to verify this. See Lemma 02QW.) We have $f^*\mathcal{F}_{\xi'} = B \otimes_A M$. Thus we see that

$$n = \text{length}_B(B \otimes_A M) \quad \text{and} \quad m = \text{length}_A(M)\text{length}_B(B/\mathfrak{m}_A B)$$

Thus the equality follows from Algebra, Lemma 02M1. □